DRISHTI

Architecture Document

Autonomous Investment Intelligence for Indian Retail Investors

ET Gen AI Hackathon 2025

5 Specialised Agents	**9 AI Routing Rules**	**12 Signal Types**	**3-Tier Fallbacks**
SSE Streaming	**Seeded Demo Mode**	**100+ Ticker Map**	**WebM Video Export**

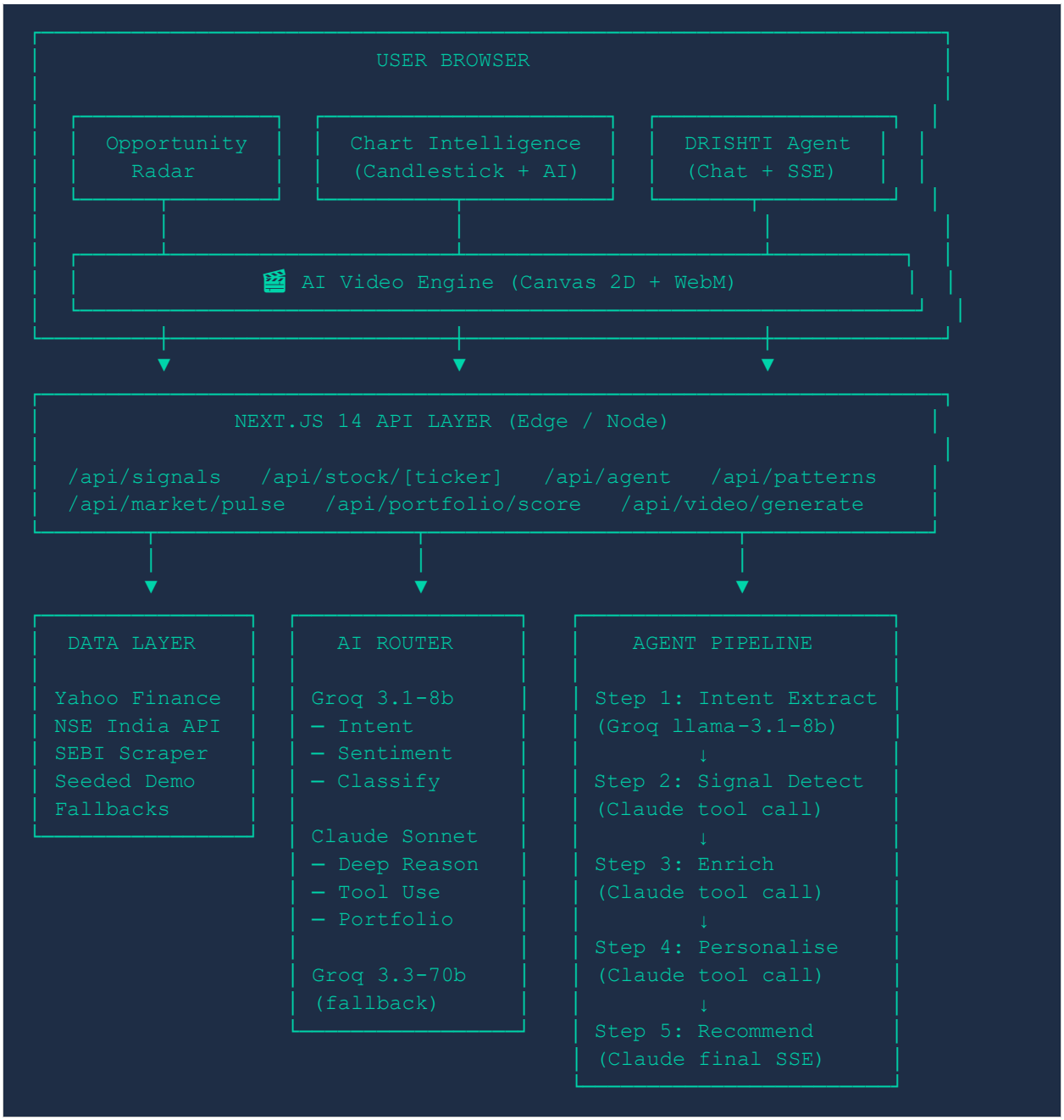
1. System Overview

DRISHTI is a multi-agent AI system built on Next.js 14 that gives Indian retail investors institutional-grade stock analysis. It routes tasks across two AI providers based on cost, latency, and capability - using Claude Sonnet for deep reasoning and Groq for fast classification and extraction.

Property	Value
Framework	Next.js 14 (App Router, SSR) - TypeScript strict mode
Primary AI	Anthropic Claude Sonnet (claude-sonnet-4-5) - deep reasoning, tool use, SSE streaming
Fast AI	Groq llama-3.1-8b-instant (extraction) + llama-3.3-70b-versatile (fallback reasoning)
Market Data	Yahoo Finance 2 (npm) + NSE India public APIs + SEBI circulars scraper
Streaming	Server-Sent Events (SSE) - no WebSocket overhead
State	Zustand + localStorage persistence
Charts	Lightweight Charts by TradingView
Video	HTML5 Canvas 2D + MediaRecorder API - zero external video libraries
Deployment	Vercel (Edge-compatible API routes)

2. High-Level Architecture

Three frontend panels communicate with the Next.js API layer. Every request passes through the AI Router, which dispatches to Claude or Groq based on task type, with deterministic seeded fallbacks at every tier.



3. Agent Roles & Responsibilities

Agent 1 - Intent Extractor	⏱ **Latency:** ~200ms	💰 **Cost:** ~$0.0001/call

Runs on every user query before Claude. Groq llama-3.1-8b-instant parses raw user input (including Hinglish and Hindi) and returns a structured object for downstream agents.

Output Field	Description	Example
ticker	NSE symbol extracted from any company name or brand	'Analyze Zomato' → ZOMATO
intent	Clean English reformulation of the query	'Reliance ka scene' → 'Analyze Reliance stock'
isHinglish	Boolean - detected language type	true if mixed Hindi/English

FALLBACK	If Groq is unavailable: 100+ entry local NAME_TO_TICKER regex map activates. Common Indian brand names (Jio, Amul, Taj Hotels, Tata Salt) pre-mapped to NSE tickers.

UNLISTED	If company not on NSE/BSE (e.g. Zerodha, boAt): Groq returns UNLISTED:<company_name>. Agent intercepts this prefix and emits a friendly explanation - prevents ₹0 data or hallucinated analysis.

Agent 2 - Stock Analyst (Primary)	⏱ **Latency:** 3–8s	💰 **Cost:** ~$0.003/call

The core reasoning agent. Claude Sonnet executes a 4-step agentic loop via registered tool use, streaming each step completion to the frontend via Server-Sent Events so users see progress in real time.

Step	Tool Used	Data Fetched	SSE Event
Signal Detection	get_stock_price, get_bulk_deals	Live price, volume ratio, NSE filings	UI Step 1
Technical Enrichment	get_technicals, get_quarterly_results	RSI, MACD, EMA20/50/200, Bollinger, ADX, ATR, quarterly PAT	UI Step 2
Fundamental Check	get_fundamentals, get_fii_dii	PE, ROE, PB, debt/equity, FII/DII flows	UI Step 3
Portfolio Personalisation	get_user_portfolio, calculate_nivesh_score	Holdings, P&L correlation, Nivesh Score (0-100)	UI Step 4

Final response is streamed as an SSE final event containing: Action badge (BUY/ACCUMULATE/WATCH/AVOID/REDUCE), entry zone, stop loss, target, Nivesh Score, Hinglish explanation, and source citations.

Agent 3 - Groq Fallback Analyst	⏱ **Latency:** ~1.5s	💰 **Cost:** ~$0.0009/call

Activates automatically when Claude API is unavailable - credit exhaustion, network timeout, or rate limit. Receives the same enriched query and full portfolio context. Uses a specialised prompt with strict output format rules (no markdown, no step narration) to produce clean verdicts in the same structure as Claude.

TRIGGER	Claude credits exhausted OR network timeout after 10s OR Claude HTTP 5xx error → Groq llama-3.3-70b receives enriched query + portfolio JSON → returns structured verdict in <3s.

Agent 4 - Pattern Detector	⏱ **Latency:** 1–6s	💰 **Cost:** 3-tier fallback

Dedicated pattern detection pipeline triggered from the Chart Intelligence panel. Receives 90-candle OHLCV data as CSV and returns a structured JSON pattern annotation.

Tier	Model	Output	Condition
Tier 1	Claude Sonnet	Full JSON: pattern name, support, resistance, target, confidence %, win rate, Hinglish explanation	Primary
Tier 2	Groq llama-3.3-70b	Same prompt, English-only output enforced, JSON format	Claude unavailable
Tier 3	Seeded Demo	12 distinct patterns, deterministic per ticker (seed = charCodeAt sum % 12). RELIANCE always gets same pattern.	Both AI unavailable

Agent 5 - Portfolio Health Scorer	⏱ **Latency:** 2–5s	💰 **Cost:** 3-tier fallback

Evaluates the user's full portfolio and returns a Health Score (0-100) with qualitative analysis and actionable improvement suggestions. Always runs the rule-based calculator in parallel as a guaranteed safety net.

Tier	Model	Output
Tier 1	Claude Sonnet	Full qualitative analysis with specific stock/sector insights and personalised suggestions

Tier	Model	Output
Tier 2	Groq llama-3.3-70b	Structured JSON: score, grade, issues array, suggestions array
Tier 3 (always parallel)	Rule-based calculator	Pure logic - always works, uses actual portfolio data, never hardcoded

Rule-based scoring logic:

- Fewer than 5 stocks in portfolio: −15 points
- Fewer than 8 stocks: −7 points
- Single sector exceeds 50% of portfolio: −12 points
- Single sector exceeds 40%: −6 points
- Single stock exceeds 35% of portfolio: −12 points
- Single stock exceeds 25%: −6 points
- Overall portfolio drawdown exceeds 15%: −10 points

4. Signals Pipeline

Endpoint: GET /api/signals - Cache TTL: 5 minutes (standard), 30 minutes (SEBI alerts). Four sources run in parallel on every refresh, results merged and capped at 12 signals.

```
┌──────────────────────────────────────────────────────────────────────┐
│                          /api/signals GET                            │
├──────────────────┬──────────────────┬──────────────────┬────────────┤
│ NSE Bulk Deals   │  Base Signals    │ Mgmt Commentary  │ SEBI Alert │
│ (live API)       │  (demo-data.ts)  │ (Groq 70b)       │ (scrape +  │
│                  │                  │                  │  Groq fb)  │
│ → getStockPrice  │ → seeded per     │ → 10 watchlist   │            │
│ → Groq sentiment │   ticker hash    │   stocks         │ 30-min TTL │
│ max 3 live deals │                  │   scanned        │            │
└──────────────────┴──────────────────┴──────────────────┴────────────┘
                                  ↓
              Merge → deduplicate → cap at 12 → cache 5min
```

Icon	Signal Type	Source	Sentiment
🔥	insider_buy	NSE corporate filings	Groq llama-3.1-8b
🚨	insider_sell	NSE corporate filings	Groq llama-3.1-8b
📈	breakout_52w	Yahoo Finance price data	Groq llama-3.1-8b
📊	volume_spike	Volume ratio > 2× 30-day average	Groq llama-3.1-8b

Icon	Signal Type	Source	Sentiment
	fii_accumulation	NSE bulk deals + FII flow data	Groq llama-3.1-8b
	fii_selling	NSE bulk deals + FII flow data	Groq llama-3.1-8b
	strong_results	NSE quarterly data (PAT > 15% YoY)	Groq llama-3.1-8b
	promoter_pledge	NSE corporate filings	Groq llama-3.1-8b
	bulk_deal	NSE bulk deal API (live)	Groq llama-3.1-8b
	reversal_pattern	Chart pattern detection engine	Groq llama-3.1-8b
	management_commentary	Groq 70b scans 10 Nifty stocks for earnings language shifts	Groq llama-3.3-70b
	regulatory_alert	SEBI circulars page (30-min cache) → Groq fallback	Rule-based

Management Commentary Signal

Groq llama-3.3-70b is prompted to surface one significant management language shift from 10 watchlist stocks: RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK, BAJFINANCE, BHARTIARTL, ZOMATO, TATAMOTORS, SUNPHARMA. Detects: guidance cuts, margin warnings, expansion announcements, CEO changes, capex plan shifts.

SEBI Regulatory Alert

- Step 1: Fetch https://www.sebi.gov.in/sebiweb/other/OtherAction.do?doRecent=yes
- Step 2: Parse circular titles from HTML table cells using regex
- Step 3: If SEBI unreachable → Groq llama-3.1-8b returns most recent notable circular from training knowledge
- Step 4: Cache result for 30 minutes to avoid hammering SEBI servers

5. AI Video Engine

Endpoint: POST /api/video/generate. A fully autonomous pipeline from raw market data to downloadable WebM video - zero human editing at any step.

```
POST body: { type: 'market_wrap' | 'race_chart' | 'fii_dii' | 'ipo_tracker' }

Pipeline:
  Step 1 → Fetch live data from /api/market/pulse (8s timeout)
           └ Fallback: getDemoMarketData() if pulse returns empty indices
  Step 2 → Groq llama-3.3-70b writes Hinglish voiceover script (max 80 words)
           └ Fallback: seeded Hinglish template per template type
```

```
  Step 3 → Return { script, marketData, sectorData, ipoData }

Frontend:
  → HTML5 Canvas 2D renders animation (requestAnimationFrame, 60fps)
  → MediaRecorder captures canvas stream at 30fps → WebM blob
  → Download URL generated → user clicks Download
  → Auto Mode: setInterval(60s) → regenerate script + re-render
```

Template	Duration	Key Scenes / Visuals
📊 Daily Market Wrap	45s	Scene 1: DRISHTI logo fade-in with glow (0-8s) | Scene 2: Index cards slide in with count-up numbers (8-20s) | Scene 3: Gainers/losers animated bar race (20-32s) | Scene 4: FII/DII arc circles (32-42s) | Scene 5: Branded outro (42-45s)
🏁 Sector Race Chart	30s	8 sectors (IT, Banking, Pharma, Auto, FMCG, Realty, Metal, Energy) racing over 5 trading days at 60fps. Leader bar highlighted in gold. Date timer top-left.
🏢 FII/DII Flow	30s	Particle stream animation between FII circle, DII circle, and NSE Market center. Green particles = buying, red = selling. Net values update live. 5-session timeline at bottom.
🚀 IPO Tracker	45s	IPO cards animate in one by one. Subscription fill bars grow to actual level. GMP badge, sector tag, status badge (oversubscribed = green, at par = yellow, undersubscribed = red).

6. Tool Integrations

Ten tools are registered with Claude via the tool_use API. Every tool has a live data source and a deterministic seeded fallback - the UI never shows ₹0, never crashes, never returns identical data for different tickers.

Tool	Data Source	Fallback
get_stock_price	Yahoo Finance 2 (TICKER.NS suffix)	TICKER_BASE_PRICES map + seeded change_pct
get_technicals	Derived from Yahoo OHLCV (RSI, EMA20/50/200, MACD, Bollinger, ADX, ATR)	Seeded per-ticker deterministic values
get_fundamentals	Yahoo quoteSummary (PE, ROE, PB, sector, debt/equity)	15-stock static dict + seeded for rest
get_bulk_deals	NSE India /api/corporates/bulk_deals (session cookie, TTL=20min)	5 realistic demo bulk deals

Tool	Data Source	Fallback
get_quarterly_results	NSE India earnings data endpoint	14-stock Q3FY26 quarterly results
get_fii_dii	Market pulse data (FII/DII net flows)	Demo FII/DII flow values
get_user_portfolio	Zustand store (passed in request body)	Empty portfolio (graceful)
calculate_nivesh_score	Internal rule engine (RSI + PE + ROE + volume ratio)	N/A - always works, pure logic
search_news	Web search via Groq tool use	Recent headline from training knowledge
get_sector_comparison	Yahoo Finance sector peers query	Static sector benchmark values

7. Data Flow - End-to-End Example

Tracing a complete request: user types "Analyze Zomato" through to final SSE response.

```
User types: "Analyze Zomato"
       │
       ▼
[Groq llama-3.1-8b]  ←── NAME_TO_TICKER local map as safety net
 Returns: { ticker: 'ZOMATO', intent: 'Analyze Zomato stock', isHinglish:
false }
       │
       ▼
[Claude Sonnet] receives: "[EXTRACTED TICKER: ZOMATO] Analyze Zomato stock"
       │
       ├─▶ tool: get_stock_price('ZOMATO')     → ₹218, +4.8%, vol 2.1×
──SSE──▶ UI Step 1
       ├─▶ tool: get_technicals('ZOMATO')      → RSI 61, MACD+, EMA bull
──SSE──▶ UI Step 2
       ├─▶ tool: get_fundamentals('ZOMATO')    → PE 312, ROE 4.2%
──SSE──▶ UI Step 3
       ├─▶ tool: get_user_portfolio()          → No ZOMATO in holdings
──SSE──▶ UI Step 4
       ├─▶ tool: calculate_nivesh_score()      → Score: 74/100
       │
       ▼
[Claude Final Response] - streamed as SSE final event:
 "ZOMATO showing breakout momentum. Technicals strong, fundamentals
stretched."
 ACTION: ACCUMULATE
 Entry: ₹212-220  |  Stop Loss: ₹198  |  Target: ₹268 (3-4 months)
 Nivesh Score: 74/100
       │
```

```
        ▼
[UI] Renders verdict with language toggle: Hindi / English / Hinglish
```

8. Error Handling & Fallback Logic

Every API route follows the same 3-tier resilience pattern. The UI is guaranteed to never show ₹0, never crash, and never return the same seeded data for different tickers.

```
TIER 1 - Live Data (Yahoo Finance / NSE API)
    │  Success → return real data
    │  Failure (timeout, 4xx, 5xx, empty response):
    ▼
TIER 2 - AI Fallback (Claude → Groq)
    │  Claude credits exhausted → Groq llama-3.3-70b
    │  Groq also fails:
    ▼
TIER 3 - Deterministic Seeded Fallback
    - Price:   TICKER_BASE_PRICES[ticker] ?? seed-derived (never ₹0)
    - OHLCV:   generateFallbackOHLCV(ticker, period) - seeded per ticker
    - Pattern: getDemoPattern(ticker) - seed = charCodeAt sum % 12
    - Score:   computeRuleBasedScore(holdings) - pure logic, always works

KEY GUARANTEE: RELIANCE always gets RELIANCE-specific seeded values.
               TCS always gets TCS-specific seeded values.
               No two tickers return identical fallback data.
```

Scenario	Behaviour
Claude API credits exhausted	Groq llama-3.3-70b takes over seamlessly - same output format
NSE bulk deal API returns 403	5 realistic demo bulk deals served from demo-data.ts
Yahoo Finance timeout (>5s)	TICKER_BASE_PRICES + seeded technicals - never ₹0
Both Claude and Groq fail	Pattern detection → seeded 12-pattern library by ticker hash
Portfolio scoring AI fails	Rule-based calculator always running in parallel as safety net
SEBI website unreachable	Groq fallback returns most recent notable circular from knowledge
Video data fetch fails	getDemoMarketData() seeded template - animation still plays
Unlisted company queried	UNLISTED:<name> prefix intercepted - friendly explanation shown

9. State Management

Zustand store with localStorage persistence. All user data (portfolio, preferences, language choice) survives page refresh.

```
Zustand Store (persisted to localStorage)
├── activeStock: string           - currently viewed NSE ticker
├── marketPulse: MarketPulseData  - indices, movers, FII/DII (polled 30s)
├── isDemoMode: boolean           - Demo vs Live mode toggle
├── portfolioOpen: boolean        - portfolio panel slide-out visibility
├── videoEngineOpen: boolean      - Video Engine overlay visibility
├── languagePreference: string    - 'hindi' | 'english' | 'hinglish'
└── portfolio
    ├── holdings: Holding[]       - stocks with qty, avg_price, date
    └── [computed by calcPortfolio()]
        ├── total_invested
        ├── current_value
        ├── total_pnl / total_pnl_pct
        └── holdings[n].pnl / pnl_pct  (per holding, live prices)
```

10. Nivesh Score Algorithm

The Nivesh Score (0-100) is DRISHTI's proprietary signal strength metric - the key differentiator between real alpha generation and news summarisation. Computed by calculate_nivesh_score() which always runs as a Claude tool call.

Factor	Condition	Points
RSI Momentum	RSI between 40-60 (healthy, not overbought/oversold)	+20
Trend Structure	Price above EMA50 (uptrend confirmed)	+15
Volume Confirmation	Volume > 1.5× 30-day average	+10
Smart Money Signal	Signal is insider_buy or fii_accumulation	+10
Fundamental Strength	Quarterly PAT growth > 15% YoY	+5
Overbought Risk	RSI > 75 (high reversal probability	−20
Downtrend Structure	Price below EMA200 (long-term downtrend)	−15
Base Score	Starting point	+50
Range	Clamped 0-100. Score > 75 = Strong. Score < 40 = Avoid.	-

11. Deployment & Environment

Variable	Purpose	Required
ANTHROPIC_API_KEY	Claude Sonnet for deep reasoning, pattern detection, portfolio scoring	Recommended
GROQ_API_KEY	Groq for intent extraction, sentiment, fallback reasoning, video scripts	Required (free tier)
NEXT_PUBLIC_DEMO_MODE	Force Demo Mode - overrides live data with seeded fallbacks	Optional

Demo Mode activates automatically when no API keys are set. All 12 signals, all chart patterns, all video animations, and all portfolio scoring work with seeded data. Only GROQ_API_KEY is needed for AI-powered natural language responses.

```
# Clone and run locally
git clone https://github.com/arnavgupta2004/Drishti.git
cd Drishti && npm install
cp .env.example .env.local
# Set ANTHROPIC_API_KEY and GROQ_API_KEY
npm run dev  →  http://localhost:3000

# Deploy to Vercel (production)
vercel --prod
# Set env vars in Vercel dashboard → Redeploy
```

DRISHTI - "दृष्टि" means Vision in Sanskrit.

Because good investing starts with seeing clearly.

github.com/arnavgupta2004/Drishti · MIT License · ET Gen AI Hackathon 2025